

UNIT III | Anatomy — Internal Structure of Plants (12 Lectures)

1.1 Plant Cell — Structure

Cell Theory: Schleiden (1838) + Schwann (1839) + Virchow (1855) | "Omnis cellula e cellula"

Organelle	Unique to Plants?	Key Function
Cell Wall	YES	Protection, shape, support; cellulose (primary) + lignin (secondary)
Chloroplast	YES	Photosynthesis; contains chlorophyll; own DNA
Large Vacuole	YES (large)	Turgor pressure, storage of pigments/waste/water
Plasma Membrane	No	Selective permeability; fluid mosaic model (Singer & Nicolson 1972)
Nucleus	No	Control centre; contains DNA; double membrane with pores
Mitochondria	No	ATP production; cellular respiration; own DNA
Ribosome	No	Protein synthesis (80S cytoplasm; 70S in plastids)
Golgi Apparatus	No	Packaging & secretion; makes cell wall pectin/hemicellulose
ER (Rough/Smooth)	No	Rough: protein synthesis; Smooth: lipid synthesis

1.2 Tissues — Classification

Tissue	Sub-type	Features	Function / Example
MERISTEMATIC	Apical	Tip of root/shoot; small cells	Primary growth (length)
MERISTEMATIC	Lateral	Vascular cambium, cork cambium	Secondary growth (girth)
MERISTEMATIC	Intercalary	Base of internode	Growth of grass internodes
Simple Permanent	Parenchyma	Living; thin walls; large vacuoles	Storage, photosynthesis, gas exchange
Simple Permanent	Collenchyma	Living; corner thickenings (cellulose)	Flexible support — celery stalk
Simple Permanent	Sclerenchyma	Dead; lignified; fibres & sclereids	Rigid support — jute, walnut shell
Complex Permanent	Xylem	Tracheids, vessels, fibres, parenchyma	Water & mineral conduction upward
Complex Permanent	Phloem	Sieve tubes, companion cells, fibres	Food conduction (source to sink)

1.3 – 1.5 Anatomy Comparison: Root, Stem, Leaf

Feature	Dicot Root	Monocot Root	Dicot Stem	Monocot Stem
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Epidermis	Epiblema with root hairs	Same; well-developed	Cuticle + stomata	Thick cuticle
Hypodermis	Absent	Absent	Collenchyma	Sclerenchyma
Vascular bundles	Radial; 2-6 (tetrarch)	Radial; polyarch (6+)	Ring; open; collateral	Scattered; closed; collateral
Cambium	Present (secondary growth)	Absent	Present (open)	Absent (closed)
Pith	Small / absent	Large and distinct	Distinct, large	Not distinct
Xylem position	Exarch	Exarch	Endarch	Endarch
Secondary growth	Present	Absent	Present	Absent

Dorsiventral (Dicot) Leaf: Upper epidermis (no stomata, thick cuticle) → Palisade mesophyll (columnar, dense chloroplasts, main photosynthesis) → Spongy mesophyll (loose, gas exchange) → Lower epidermis (stomata abundant, kidney-shaped guard cells).

Isobilateral (Monocot) Leaf: Both epidermis have stomata (amphistomatic); mesophyll NOT differentiated; Kranz anatomy (bundle sheath) for C4 photosynthesis; bulliform cells help in leaf rolling during drought.